Ram Certificates for Turn-Based Stochastic Mean-Payoff Games

Quantitative Discounting, Exact Zero, Behavioral Opponents,
and the Quasipolynomial Labeling Interface

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Abstract

We examine the complete mathematical source of the pinned preprint *Turn-Based Stochastic Mean-Payoff Games in Deterministic Quasipolynomial Time*. Its asserted algorithm decides the nonnegative expectation-of-pathwise-liminf value set for explicit finite turn-based games with binary integer rewards and rational probabilities. We rederive the rational discount and separation estimates, explain the simultaneous-labeling and bit-complexity chain, and separate them from locally proved certificate results. The latter include componentwise discounted residual envelopes, asymmetric policy repairs against all behavioral opponents, exact-zero tests using the rational gap, and a finite gain–bias certificate with a quantitative bound on gain changes. We also give a sharper reachability-to-discount comparison under a supplied stopping drift. These results retain the physical state space and all residual rows; they do not turn a small nonstopping Bellman residual into a value certificate or replace policy discovery by linear algebra. Classical methods and the actual private residual/stopping passages are explicitly credited. Finite exact tests support bounded identities, not a new universal solver or a measured speedup. Contribution assessment: C2/5, source-specific certificate refinements and complete bounded proofs without a new global game-solving complexity result.

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1 Scope, Provenance, and Claim Discipline

The target is the OpenAI preprint dated 5 October 2026 [1], at public repository pin

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The actual mathematical input is `build/build/main.tex`, its four sections `introduction.tex`, `discount.tex`, `labeling.tex`, and `application.tex`, together with `figures/labeling.tex` and the bibliography. All these files were read, not merely their abstracts or catalogue descriptions. The report concerns the exact model below, not simultaneous stochastic games, partial observation, or succinct arenas.

Status	Meaning
Source claim	A deterministic $2^{O((\log(L+2))^2)}$ bit algorithm for the nonnegative-value set, including zero.
Classical input	Contraction, rational-function sign stabilization, finite-state martingale arguments, nonnegative resolvents, rational LP, and determinant height bounds.
Local proof	Residual and policy certificates, a gain–bias verification theorem, exact-zero margins, and a stopping-sensitive discount bound.
Conditional bridge	A supplied accurate reconstruction, exact gain inequalities, or a stopping drift is checked before using its conclusion.
Not established	A new faster discovery algorithm, a polynomial all-game solver, an empirical speedup, or historical novelty of these classical tools.

Applicability L4 records the close match of the residual and finite-game interfaces. Contribution C2 is a different scale: the bounded refinements proved here do not constitute the source's global quasipolynomial algorithm.

1.1 Actual private mathematical input

The locally supplied `TR-item-312-astra.tex` [2] was read at its actual finite-domain, policy-evaluation, nonstopping-certificate, residual-envelope, polynomial-duration, counterexample, and finite-synthesis passages (source lines 123–771). In its notation, the reachability operator acts on $\mathbb{R}^V$, with target value one, failure value zero, and nontermination payoff zero. A common rational drift satisfies

$$h_i \geq 1 + \sum_{j \in V} P^a_{ij} h_j, \qquad h \geq \mathbf{1}, \quad \text{for every state and every available row } a. \tag{1.1}$$

Its envelope theorem checks $e_i \geq d_i + \sum_j P^a_{ij} e_j$ for every row and concludes $|x - v| \leq e$ only under the common stopping hypothesis. Its lower-policy certificate instead fixes Max's policy and checks drift only on the proposed positive region, against every Min action. Its upper certificate requires a nonnegative supersolution but no stopping promise. These are distinct quantifiers and are not interchanged here.

The same manuscript explicitly distinguishes polynomial numeric duration from polynomial encoding length, and transports finite synthesis through the full physical residual. Those actual results, not an assumed universal Ram inverse, motivate Sections 5–6 below. No theorem from the foundational reports cited inside that manuscript is claimed as independently read or imported here. All confidential passages, paths and hashes remain local; public checks use public titles and URLs only.

2 The Exact Game and Reduction

Let $V = \{1, \dots, n\}$, $n \geq 1$, partition into Max, Min and chance vertices. The explicitly listed finite directed multigraph has at least one outgoing edge per vertex, integer edge rewards $w(e)$, and rational chance probabilities $p_i(e) \geq 0$ with sum one at each chance vertex. Loops, parallel edges, zero probabilities and unreduced input fractions are allowed. The full binary input length L counts the graph, ownership, rewards, numerators and positive denominators. Every strategy can depend on the complete observed finite history and use private randomness. Define

$$\mathrm{MP}_w = \liminf_{T\to\infty} \frac{1}{T} \sum_{t<T} w(e_t), \qquad \mathrm{val}(i) = \sup_{\sigma} \inf_{\tau} \mathbb{E}_{i,\sigma,\tau}\, \mathrm{MP}_w\,. \tag{2.1}$$

Expectation is outside the pathwise liminf. There is no interchange of these operations in this report. The source computes $\{i : \mathrm{val}(i) \geq 0\}$, not necessarily a printed optimal policy pair or exact rational value vector.

Proposition 2.1 (Reachability, including nontermination)**.** *In a finite turn-based reachability game, make the target absorbing with reward $+1$, and put reward -1 on all other edges, adding failure loops where necessary. For every behavioral strategy pair, $\mathrm{MP}_w = 2\mathbf{1}_{\{\text{target reached}\}} - 1$ pathwise. Hence the new mean-payoff value is $2v^{\mathrm{reach}} - 1$ and its nonnegative set is exactly the reachability threshold-$1/2$ set.*

Proof. If the target is reached at finite time, all but finitely many rewards are $+1$; otherwise all rewards are -1. This includes a play that never reaches any terminal. Taking expectation and then infimum and supremum commutes with the increasing affine map $r \mapsto 2r - 1$. The construction adds at most a linear number of explicitly encoded loops and unit rewards. □

The proof does not need a stopping hypothesis, but the quantitative comparison with a fixed positive discount generally does.

3 Quantitative Discount Algebra

For $0 < u < 1$ put $\gamma = 1 - u$ and define $f_u : \mathbb{R}^V \to \mathbb{R}^V$ by taking Max's maximum, Min's minimum, or the prescribed chance average of $uw(e) + \gamma t_j$ on edges $e : i \to j$. It is monotone, γ-contractive in $\|\cdot\|_\infty$, and satisfies $f_u(t + c\mathbf{1}) = f_u(t) + \gamma c\mathbf{1}$. Its unique fixed point $v(u)$ lies in $[-W, W]^V$, where $W = \max(1, \max_e |w(e)|)$. The source uses

$$\begin{aligned} Q &= \prod_{i\in V_{\mathrm{ch}}} \prod_{e \text{ out of } i} \mathrm{den}(p_i(e)), \qquad A_0 = n!\,[Q(3+W)]^n, \\ H &= (W+4)A_0, \qquad \lambda_0 = (32H^2)^{-1}. \end{aligned} \tag{3.1}$$

Here den denotes the supplied positive denominator, and an empty product is one. We use this unreduced product rather than quietly assuming a common small denominator. Since $n = O(L)$ and $\log Q, \log W = O(L)$, $\log H = O((L+2)^2)$.

3.1 Fixed-pair polynomials and a uniform sign interval

Fix a pure positional pair, aggregate chance transitions into a stochastic matrix P, and put $r_i = \mathbb{E}[w(e) \mid i]$. Different parallel-edge rewards are averaged in r, not lost when successors are merged.

Its discounted value solves $x = ur + (1-u)Px$. Invertibility follows because a kernel vector would satisfy $\|x\|_\infty \le (1-u)\|x\|_\infty$. Cramer's rule gives

$$x_i(u) = N_i(u)/D(u), \qquad D(u) = \det\big(Q(I-(1-u)P)\big). \tag{3.2}$$

These are integer polynomials. With coefficient ℓ^1 norm, determinant expansion gives $\|N_i\|_{\mathrm{coef}}, \|D\|_{\mathrm{coef}} \le A_0$: ordinary entries have norm at most $3Q$, and replacing-column entries Qur_i have norm at most QW. An edge-optimality numerator is

$$N_i - uw(e)D - (1-u)N_j,$$

whose coefficient norm is at most $(W+3)A_0 \le H$. For any nonzero integer polynomial of norm at most H, write $a(u) = u^k(c + uq(u))$ with nonzero integer c. On $0 < u \le 1/(2H)$, $|uq(u)| \le Hu \le 1/2 < |c|$. Its sign is constant there. The denominator has constant nonzero sign as well. Selecting attaining edges at $v(\lambda_0)$ therefore supplies a single pair whose inequalities hold for every $0 < u \le 1/(2H)$; uniqueness makes its solution $v(u)$.

3.2 Gain, bias, and the gap

If u^k is the exact zero factor of D, boundedness of every N_i/D implies divisibility of N_i by u^k. Set $\widehat{D} = D/u^k$ and $\widehat{N}_i = N_i/u^k$. The quotient is analytic at zero, and

$$v_i(u) = G_i + u\psi_i + O(u^2), \quad G_i = \widehat{N}_i(0)/\widehat{D}(0), \quad |G_i| \le W. \tag{3.3}$$

The nonzero integer denominator has magnitude at most A_0, so $G_i \neq 0$ implies $|G_i| \ge 1/A_0 \ge 1/H$. Since $\widehat{N}_i - G_i\widehat{D}$ has zero constant coefficient and norm at most $(1+W)A_0$, while $|\widehat{D}(u)| \ge 1 - A_0u \ge 1/2$,

$$|v_i(u) - G_i| \le 2(1+W)A_0u \le 2Hu \quad (0 < u \le 1/(2H)). \tag{3.4}$$

In particular the error at λ_0 is at most $1/(16H)$. The denominator and error estimates alone do not identify G with (2.1); the behavioral-opponent proof is supplied in Section 4.

4 Gain–Bias Certificates Against Behavioral Opponents

The following is a certificate formulation of the source's probabilistic argument, extended to a controlled bias defect. It is not a new stationary optimality theorem or an algorithm for finding the certificate.

Lemma 4.1 (Finite-valued drift with a change-count bound)**.** *Let an adapted process Z_t take values in a fixed finite set and satisfy $\mathbb{E}[Z_{t+1} \mid \mathcal{F}_t] \ge Z_t$. Let R be the difference of its largest and smallest possible values. If $R > 0$, let $\Delta > 0$ be the least distance between distinct values. The number C of changes obeys*

$$\mathbb{E}C \le R^2/\Delta^2.$$

In particular Z_t is eventually constant almost surely and $\mathbb{E}Z_\infty \ge \mathbb{E}Z_0$. If $R = 0$, there are no changes. The same conclusions with the expectation inequality reversed hold for a supermartingale.

Proof. Shift by the least possible value, writing $0 \le Y_t \le R$. Then

$$\mathbb{E}(Y_{t+1}^2 - Y_t^2) = \mathbb{E}(Y_{t+1} - Y_t)^2 + 2\mathbb{E}\big[Y_t\mathbb{E}(Y_{t+1} - Y_t \mid \mathcal{F}_t)\big] \ge \mathbb{E}(Y_{t+1} - Y_t)^2.$$

Summation and monotone convergence give $\mathbb{E}\sum_{t\ge0}(Y_{t+1} - Y_t)^2 \le R^2 - \mathbb{E}Y_0^2 \le R^2$. Each change contributes at least Δ^2. Hence C has finite expectation and is finite almost surely. Bounded convergence proves the claim for Z_∞. Negation treats the supermartingale case. □

This quantitative bound counts changes, not elapsed time to the last change. An arbitrarily rare transition can give a long waiting time with only one change; the bound must not be used as a polynomial stopping-time estimate.

Lemma 4.2 (Bounded martingale averages)**.** *If D_t is $\mathcal{F}_{t+1}$-measurable, $|D_t| \le c < \infty$, and $\mathbb{E}[D_t \mid \mathcal{F}_t] = 0$, then $T^{-1}\sum_{t<T} D_t \to 0$ almost surely.*

Proof. Distinct increments are orthogonal in L^2, so the variance of their sum is at most Tc^2. At $T = k^2$, Chebyshev's inequality bounds the probability of an average exceeding ε in absolute value by $c^2/(\varepsilon^2 k^2)$. Summability and the union bound on tails show almost-sure convergence along squares, simultaneously for rational positive ε. The at most $2k+1$ remaining bounded terms between successive squares have average tending to zero. This proves the claim for all T. □

Theorem 4.3 (Robust finite lower and upper certificates)**.** *Supply $g, b \in \mathbb{R}^V$, $\varepsilon \ge 0$, and a pure positional Max policy σ. Require the following finite tests:*

1. *On the selected Max edge $i \to j$, $g_j = g_i$ and $w(e) + b_j - b_i \ge g_i - \varepsilon$.*
2. *On every Min edge $i \to j$, $g_j \ge g_i$; whenever equality holds, $w(e) + b_j - b_i \ge g_i - \varepsilon$.*
3. *At each chance vertex, $\sum_e p_i(e) g_j = g_i$ and $\sum_e p_i(e)(w(e) + b_j) - b_i \ge g_i - \varepsilon$.*

Then for every start i and every behavioral Min strategy τ,

$$\mathbb{E}_{i,\sigma,\tau}\,\mathrm{MP}_w \ge g_i - \varepsilon.$$

An upper certificate interchanges the players and reverses the inequalities: a selected Min edge preserves g and has bias expression at most $g_i + \varepsilon$; every Max edge weakly decreases g, with that bias bound on gain-preserving edges; chance gain is harmonic and its expected bias expression is at most $g_i + \varepsilon$. It guarantees $\mathbb{E}_{i,\sigma,\tau}\,\mathrm{MP}_w \le g_i + \varepsilon$ for every behavioral Max opponent. Separate lower and upper vectors may be supplied.

Proof. For the lower certificate, $g(X_t)$ is a finite-valued submartingale in the filtration of the full observed history. Randomized Min actions preserve the drift inequality by averaging. Lemma 4.1 gives eventual constant g_∞ and $\mathbb{E}g_\infty \ge g_i$. Put $R_t = w(e_t) + b(X_{t+1}) - b(X_t) - g(X_t)$. At chance vertices let $m_t = \mathbb{E}[R_t \mid \mathcal{F}_t]$ and set $D_t = \mathbf{1}_{\{X_t \in V_{\mathrm{ch}}\}}(R_t - m_t)$, with $D_t = 0$ elsewhere. These are bounded martingale differences, and $m_t \ge -\varepsilon$ at chance vertices. After the last gain change, every traversed player edge preserves gain; the player residuals are therefore at least $-\varepsilon$. Thus, outside a finite initial segment, $R_t - D_t \ge -\varepsilon$ at every step. Lemma 4.2 and telescoping give

$$\frac{1}{T}\sum_{t<T} w(e_t) = \frac{1}{T}\sum_{t<T} g(X_t) + \frac{1}{T}\sum_{t<T} R_t + \frac{b(X_0) - b(X_T)}{T}, \tag{4.1}$$

so the liminf is at least $g_\infty - \varepsilon$ almost surely. Taking expectations is legitimate since rewards and gains are bounded. For the upper certificate $g(X_t)$ is a supermartingale and the same argument bounds the limsup by $g_\infty + \varepsilon$. The liminf is no larger. Its expectation is consequently at most $g_i + \varepsilon$. □

Corollary 4.4 (Source gain identification and exact witnesses)**.** *The G, ψ of (3.3), with the source's small-discount pair, pass both tests at $\varepsilon = 0$. Hence $G = \mathrm{val}$ and the pair is optimal against every behavioral opponent for (2.1). For rational submitted data, all tests in Theorem 4.3 have polynomial bit cost in input plus certificate length.*

Proof. Substitute (3.3) into each selected discounted equation. The constant and linear coefficients give $G_i = G_j$ and $\psi_i = w(e) + \psi_j - G_j$. At chance vertices they give $G_i = \sum pG_j$ and $\psi_i = \sum p(w + \psi_j - G_j)$, hence the required expected bias equality. A Min edge inequality first gives $G_i \le G_j$. When these are equal, subtract the constant, divide by u, and take the limit to obtain $\psi_i \le w + \psi_j - G_j$. Max inequalities reverse. Thus the theorem gives matching guarantees, proving the value identity. Finite rational dot products, exact gain equalities, edge comparisons and selected-edge tests take polynomial time in their supplied binary lengths. □

Polynomial-size exact certificates exist here: G has denominator at most A_0, and if n_0, n_1, d_0, d_1 are the first two coefficients of the cancelled Cramer polynomials, then $\psi_i = (n_1 d_0 - n_0 d_1)/d_0^2$. Each coefficient has magnitude at most A_0, so these rationals have $O(\log A_0)$ bits. This existence does not make discovery polynomial. The new certificate formulation tolerates bounded bias defects but deliberately requires exact gain drift. Replacing those exact conditions by small violations is unsound over infinite time. For a concrete counterexample, a reward-zero chance state escapes with probability $p > 0$ to a reward-zero absorbing state. Propose $g = (1, 0)$, $b = (-1/p, 0)$. The chance bias identity holds exactly, but its gain drift is $-p$. Both true values are zero, so allowing this arbitrarily small negative drift would falsely certify a unit lower value. Large bias is allowed in the theorem; it is exact gain harmonicity that excludes this counterexample.

5 Discounted Physical Residual Certificates

For each player edge use a deterministic row P_i^a and reward $r_i^a = w(e)$; at chance vertices use the single aggregated stochastic row and expected reward. Write

$$B_i^a(x) = ur_i^a + \gamma P_i^a x, \qquad f_u(x)_i = \operatorname*{ext}_a B_i^a(x),$$

where the extremum is maximum or minimum as appropriate and is the identity on the one chance row. The following bounds are on the full space $\mathbb{R}^V$, not merely a coefficient subspace or the reward box.

Theorem 5.1 (Componentwise discounted envelope)**.** *Fix $0 < u < 1$, $x \in \mathbb{R}^V$, $d \ge |f_u(x) - x|$, and $e \ge 0$ such that*

$$e_i \ge d_i + \gamma P_i^a e \qquad \text{for every state and every row } a. \tag{5.1}$$

Then $|v(u) - x| \le e$. The coordinatewise least feasible envelope is

$$e_i^* = \sup_\pi \mathbb{E}_i^\pi \sum_{t \ge 0} \gamma^t d(X_t), \tag{5.2}$$

where a single controller chooses all rows. It uniquely minimizes $\sum_i e_i$ subject to (5.1) and $e \ge 0$. For rational u, d, P, this is a rational LP of polynomial bit complexity in their combined encoding length. Always $e^* \le (\|d\|_\infty / u)\mathbf{1}$.

Proof. Finite extrema are Lipschitz in the largest rowwise change. Therefore

$$f_u(x + e)_i \le f_u(x)_i + \gamma \max_a P_i^a e \le x_i + e_i,$$

and $f_u(x - e)_i \ge x_i - e_i$ by the reverse inequality. Iteration of these sub- and supersolutions converges by contraction to $v(u)$. The map $E(z)_i = d_i + \gamma \max_a P_i^a z$ is a contraction. Its iterates from zero increase to its unique nonnegative fixed point. Every feasible e dominates every iterate and hence

that limit; the limit is itself feasible. Finite-horizon dynamic programming identifies the iterates with maximum discounted defect costs. Passing to the increasing limit gives (5.2): each policy's infinite cost is the limit of its finite costs, while a maximizing stationary selection at the limit attains it, since the bounded discounted terminal remainder tends to zero. Coordinatewise minimality makes it the unique sum minimizer. The constant vector $\|d\|_\infty / u$ is feasible because $1 - \gamma = u$. The constraints are explicitly listed rational linear inequalities of polynomial size; polynomial-time rational LP is the classical imported algorithm, not a mixed-game Bellman solver. □

This specializes the private full-row envelope principle to automatic geometric killing. The factor $1/u$ can be numerically exponential while having only polynomially many bits. Computing a defect envelope after x is supplied is not computing a sufficiently accurate x for free.

Theorem 5.2 (Asymmetric policy repair)**.** *Fix a Max policy σ and keep only its row at Max states, all Min rows, and the single chance row; denote these sets by $\mathcal{A}_i^\sigma$. For arbitrary $x \in \mathbb{R}^V$, take $d^- \geq 0$, $e^- \geq 0$ satisfying*

$$d_i^- \geq x_i - B_i^a(x), \qquad e_i^- \geq d_i^- + \gamma P_i^a e^- \quad (a \in \mathcal{A}_i^\sigma). \tag{5.3}$$

Then σ guarantees discounted payoff at least $x_i - e_i^-$ against every behavioral Min strategy, and $v_i(u) \geq x_i - e_i^-$. For an upper repair, fix Min's policy, retain all Max rows, require $d_i^+ \geq B_i^a(x) - x_i$ and the analogous envelope. Then that Min policy guarantees at most $x_i + e_i^+$ against every behavioral Max strategy.

Proof. Let $\ell = x - e^-$. For every retained row,

$$\ell_i - B_i^a(\ell) = x_i - B_i^a(x) - e_i^- + \gamma P_i^a e^- \leq 0.$$

Conditional on any observed history under σ, this implies $\ell(X_t) \leq \mathbb{E}[uw(e_t) + \gamma \ell(X_{t+1}) \mid \mathcal{F}_t]$. Multiply by γ^t, sum, and take expectations to get $\ell_i \leq \mathbb{E} \sum_{t<T} u\gamma^t w(e_t) + \gamma^T \mathbb{E}\ell(X_T)$. The last term vanishes since ℓ is bounded on finite V. This proves the lower guarantee for the normalized discounted payoff. Reverse all inequalities for the upper repair. Selecting attaining edges at $v(u)$ gives matching exact discounted guarantees by the same telescoping proof, so this payoff's value is indeed the fixed point used above. □

The repairs use only the necessary opponent rows and one-sided defects; they may be strictly smaller than a universal absolute-residual envelope. They certify the displayed policies only for the discounted game. A mean-payoff guarantee for those same policies needs the gain–bias tests, not just closeness of the discounted value to G.

5.1 Exact zero and adaptive certificate margins

Proposition 5.3 (Gap-aware mean-payoff classification)**.** *At $u = \lambda_0$, let $[\ell_i, r_i]$ be any certified interval for $v_i(u)$ and put $a_i = \ell_i - 1/(16H)$, $b_i = r_i + 1/(16H)$. Then $G_i \in [a_i, b_i]$. The following exact rational tests are sound:*

1. *$a_i > -1/H$ implies $G_i \geq 0$; $b_i < 0$ implies $G_i < 0$.*
2. *$a_i > -1/H$ and $b_i < 1/H$ together imply $G_i = 0$.*

In particular if $|x_i - v_i(u)| \leq e_i$ and $e_i + 1/(16H) \leq 1/(8H)$, then $G_i \geq 0$ if and only if $x_i \geq -1/(2H)$.

Proof. Equation (3.4) gives the interval containment and Corollary 4.4 identifies G with the game value. Every negative G_i is at most $-1/H$; every positive G_i is at least $1/H$. Exclusion of these ranges proves the first two tests, including the strict endpoints. In the final assertion $G_i \geq 0$ implies $x_i \geq -1/(8H) > -1/(2H)$; $G_i < 0$ implies $x_i \leq -7/(8H) < -1/(2H)$. These cases are exhaustive. □

The equality conclusion uses the independently proved rational gap, not the heuristic that a small computed number must equal zero. If a current interval meets a gap endpoint, the stated strict test simply declines to classify it. No invalid answer is forced.

5.2 Strict action exclusion and coordinate transport

For a certified $|v(u) - x| \leq e$, set $L_i^a = B_i^a(x) - \gamma P_i^a e$ and $U_i^a = B_i^a(x) + \gamma P_i^a e$. At Max, $U_i^b < L_i^a$ proves that b cannot attain the discounted maximum; at Min, $L_i^b > U_i^a$ excludes b from the minimum. Indeed each true $B_i^a(v(u))$ lies in its interval. Strictly excluded actions can be removed while retaining an attaining pair, so the discounted fixed point is unchanged. At $u = \lambda_0$, every fully attaining original pair stabilizes by the sign argument of Section 3 and is mean-payoff optimal. This does not assert that a removed action was never mean-payoff optimal; Blackwell optimality is a stronger selection criterion.

For a fixed pair, set $A_u = I - \gamma P$ and choose a rational synthesis $S \in \mathbb{Q}^{n \times d}$ of full column rank. For any coefficient vector c,

$$r = ur^{\text{pair}} - A_u Sc, \qquad x^{\text{pair}}(u) - Sc = A_u^{-1} r, \qquad |x^{\text{pair}}(u) - Sc| \leq A_u^{-1}|r|. \tag{5.4}$$

The inverse is the nonnegative series $\sum_{k \geq 0} \gamma^k P^k$; multiplication and contraction prove the identities. Rectangular synthesis does not remove the residual in omitted physical directions. A test matrix W_0 with $W_0 r = 0$ need not imply $r = 0$: take $A_u = uI$, $r^{\text{pair}} = (0,1)^T$, $Sc = 0$, and $W_0 = (1,0)$. The measured residual vanishes but the second-state error equals one. For square invertible S, coefficient order must be $c \preceq_S d \iff Sc \leq Sd$, and the norm must be $\|c\|_{S,\infty} = \|Sc\|_\infty$. A negative coefficient is not a negative physical probability or reward value. Similarity preserves eigenvalues, so it cannot uniformly enlarge a vanishing spectral gap merely by renaming coordinates.

6 A Stopping-Sensitive Reachability Bridge

The private reachability envelope is not directly a theorem about signed expected-liminf rewards. Proposition 2.1 supplies the exact semantic link; the next result supplies a conditional quantitative link.

Theorem 6.1 (Discount error from a common absorption drift)**.** *For the reachability game of Proposition 2.1, suppose a rational drift $h \geq \mathbf{1}$ on nonterminals satisfies (1.1), with zero extension on true target/failure terminals. Let $u \in (0,1)$ and $v(u)$ be the normalized discounted value of the signed reward construction. Then for every starting nonterminal i,*

$$0 \leq (2v_i^{\text{reach}} - 1) - v_i(u) \leq 2uh_i. \tag{6.1}$$

More precisely, if τ_F is target hitting time, with $\gamma^\infty \mathbf{1}_{\{\tau_F < \infty\}} = 0$, the discounted payoff for every strategy pair has expectation $2\mathbb{E}[\gamma^{\tau_F}\mathbf{1}_{\{\tau_F<\infty\}}] - 1$.

Proof. The first τ_F edges pay -1 and subsequent edges pay $+1$, so the normalized geometric sum is $2\gamma^{\tau_F} - 1$ on target-reaching paths and -1 otherwise. The loss from the mean payoff is therefore

$2(1-\gamma^{\tau_F})\mathbf{1}_{\{\tau_F<\infty\}} \geq 0$. Let ρ be first hitting time of either true terminal. Summing the conditional drift for $h(X_{t\wedge\rho})$ gives $\mathbb{E}(t \wedge \rho) \leq h_i$, hence $\mathbb{E}\rho \leq h_i$. On target-reaching paths $\tau_F = \rho$, and $1-(1-u)^k \leq uk$ for every integer $k \geq 0$. The expected loss is at most $2u\mathbb{E}[\rho\mathbf{1}_{\{X_\rho\in F\}}] \leq 2uh_i$. These uniform inequalities hold for every behavioral pair at start i; taking infima and suprema preserves the same additive constant and proves (6.1). □

The improvement is instance-sensitive: a supplied small h_i may replace the large universal H in a reachability error budget. A common drift with polynomially bounded *numeric* entries permits duration-based polynomial iteration, as in the private manuscript. A drift whose entries only have polynomial bit length need not. No such drift exists for a nonterminal closed class, and no arbitrary positive discount preserves all exact threshold answers without an explicit separation margin.

6.1 Sharp obstruction families

For one chance state with escape probability $p = 2^{-b}$ to target and self-loop probability $1-p$, reachability value is one, $h = 1/p$, and $T(0) - 0 = p$. Unit error coexists with exponentially small residual. In the signed construction the target loop has reward $+1$, and every pre-target edge has reward -1. Direct solution gives

$$v_s(u) = 2\frac{\gamma p}{u+\gamma p} - 1, \qquad 1 - v_s(u) = \frac{2u}{u+\gamma p}. \tag{6.2}$$

For fixed p and $u \downarrow 0$, the ratio of this error to $2u/p$ tends to one. Thus the coefficient two in (6.1) is asymptotically sharp. At $u = p$, $v_s(u) = -p/(2-p) < 0$ although the mean-payoff value is $+1$. The discount choice is a substantive obligation.

In an undiscounted Max reachability state choosing a self-loop or an exit of success probability $1/2$, $T(x) = \max(x, 1/2)$. Every $x \in [1/2, 1]$ has zero residual, but the true value is $1/2$, and the greedy self-loop has value zero. A Min choice of self-loop or success gives $T(x) = x$ on $[0, 1]$ and true value zero. These are failures of unqualified residual inference, not lower bounds against all algorithms. The scalar rare-escape equation itself is easy to solve exactly.

6.2 When the common drift is only a promise

With $H_h = \max h_i$ and $\|z\|_h = \max_i |z_i|/h_i$, the private operator satisfies $\|Tx - Ty\|_h \leq (1 - H_h^{-1})\|x - y\|_h$ because every row sends h to at most $h - \mathbf{1}$. Rounding each iterate to absolute error η and clipping to $[0, 1]^V$ gives

$$\|y_t - v\|_\infty \leq H_h(1 - H_h^{-1})^t + \eta H_h^2.$$

The induction uses a weighted one-step rounding error at most η, a geometric series at most H_h, and conversion back to sup norm by another factor H_h. Taking $\eta \leq \epsilon/(2H_h^2)$ and $t \geq H_h \log(2H_h/\epsilon)$ suffices, with the $H_h = 1$ case handled in one step. Only a polynomial numeric bound on H_h makes this a polynomial bit algorithm at binary precision. Rational reconstruction additionally needs a proved denominator bound; it is not licensed by numerical convergence alone.

7 The Source's Simultaneous Labeling Chain

This section explains the full source-specific logical chain. The recursive procedure and complexity claim belong to the source, not to the private residual manuscript or the local envelope improvements. The abstract input is a monotone $F : \mathbb{R}^n \to \mathbb{R}^n$ with

$$F(t + c\mathbf{1}) = F(t) + \gamma c\mathbf{1} \quad (t \in \mathbb{R}^n, c \in \mathbb{R}), \qquad 0 < \gamma < 1. \tag{7.1}$$

Its oracle returns exact $\lfloor F(t) \rfloor$ for integer t. A box has lower corner $A \in \mathbb{Z}^n$, width $D = h2^d$ with $128 \le h \le 255$, $d \ge 0$, upper corner $U = A + D\mathbf{1}$, strip width $P_0 = D/32$, and positive rational mass vectors a, b.

7.1 Quantified witnesses and comparison

An integer $y \in [A, U]$ is a subsolution if $F_i(y) \ge y_i$ whenever $y_i > A_i$. It is a plus witness when $a(\{i : y_i > A_i\}) \le 1$, and claims i if $y_i \ge U_i - P_0$. An integer $z \in [A, U]$ is a supersolution if $F_i(z) \le z_i$ whenever $z_i < U_i$; it is a minus witness when $b(\{i : z_i < U_i\}) \le 1$, claiming i if $z_i \le A_i + P_0$. The output must respect *every* such claim, simultaneously, without being given the witnesses.

For any subsolution y and supersolution z, a positive maximum $m = \max_i(y_i - z_i)$ at i would activate both inequalities and give $m \le F_i(y) - F_i(z) \le \gamma m$, a contradiction. Thus $y \le z$. The clipped-floor iteration

$$t^0 = U, \qquad t^{k+1} = \min\{U, \max\{A, \lfloor F(t^k) \rfloor\}\}$$

decreases in the finite integer box and stops in at most $nD+1$ evaluations. Each subsolution is below every iterate; its terminal fixed point t^* is itself a subsolution and hence below every supersolution. This gives correct labels by midpoint comparison when $d < 12$, because $D \le 522240$ is then an absolute constant. For large D, performing nD iterations would destroy the desired bit bound.

7.2 Two pivot passes and the signed translations

For nonbase calls let $c = A + (D/2)\mathbf{1}$, $M = D/8$, $q = D/128$, $p = D/4096$. The pivot stays in $[c - M\mathbf{1}, c + M\mathbf{1}]$. A centered width-$r$ child has box $[k - (r/2)\mathbf{1}, k + (r/2)\mathbf{1}]$, where $r = q$ or $D/2$. For a relevant plus witness define $s = \max_i(y_i - k_i)$ and $g_i = s - (y_i - k_i)$, and translate and clip by

$$y'_i = \max\{k_i - r/2, y_i - s + r/2\}.$$

A claim ensures $s \ge 11D/32$, so $t = -s + r/2 < 0$. On the surviving support $\{g_i < r\}$, the old subsolution inequality applies and $F_i(y') \ge F_i(y) + \gamma t \ge y_i + t = y'_i$. Outside the parent support the gap is at least $23D/32 > r$, so no new support appears. Every $g_i \le r/32$ becomes a claim. For minus witnesses use $s = \max_i(k_i - z_i)$ and $z'_i = \min\{k_i + r/2, z_i + s - r/2\}$. Now $t = s - r/2 > 0$ and $\gamma t \le t$, preserving the supersolution inequality. The sign of each translation, not just formal translation invariance, is essential for discounted maps.

The downward pass starts at $k = c$, calls centered width q with masses $(a, 4b/3)$, and simultaneously moves minus-labeled coordinates down by p, clipped at $c_i - M$. It stops at an unchanged update and defines $I_- = \{i : k_i^{\downarrow} < c_i - P_0\}$. Because every plus gap at most p must receive plus, all plus peaks remain fixed; other coordinates cannot overtake the peak in a step of size p. Thus no plus witness claims in I_-. A minus witness claiming outside I_- has peak at least $7D/16$, strictly above the largest possible capped deviation $D/2 - M$. If its near-peak support $\{g_i < q\}$ had b-mass at most $3/4$, its translated witness would be admissible for the final child and force an uncapped peak coordinate to move. Contradiction. Its mass is therefore strictly greater than $3/4$.

The upward pass starts at $k^{\downarrow}$, calls centered width q with $(4a/3, b)$, and moves plus-labeled coordinates up by p, clipped at $c_i + M$. At its final pivot k^f define $I_+ = \{i : k_i^f > c_i + P_0\}$. The reversed argument preserves minus peaks and excludes minus claims from I_+. Minus gaps can only decrease, so the first pass's minus mass bound survives. The stopped second pass forces plus near-peak mass greater than $3/4$ for every plus witness claiming outside I_+. Each pass has at most $1024n + 1$ child calls, including its unchanged final call: a coordinate travels at most $2M$ in increments p, with $2M/p = 1024$.

7.3 Reweighting, overrides, and rank

A provisional centered width-$D/2$ call at k^f uses unchanged masses. Since $q = D/128 < (D/2)/32$, all near-peak coordinates just found are claims in this larger child. For provisional label plus multiply (a_i, b_i) by $(4/5, 8/5)$, and for minus by $(8/5, 4/5)$. For a plus witness with support S and a claim outside I_+, its new support mass is

$$\bar{a}(S) = \tfrac{8}{5}a(S) - \tfrac{4}{5}a(S \cap Q_+) < \tfrac{8}{5} - \tfrac{4}{5} \cdot \tfrac{3}{4} = 1.$$

The minus calculation is identical with b, Q_-. Therefore these witnesses remain admissible for a continuation call in the unchanged box with new masses. Finally force minus on I_- and plus on $I_+ \setminus I_-$. Plus claims never lie in I_-; minus claims never lie in I_+. Thus even overlapping override sets cannot corrupt a claim.

Set $\rho_0 = 32/25$, $\mu = \min_i a_i b_i$, and $B(\mu) = \min\{j \geq 0 : \rho_0^j \mu > 1\}$. In a mass base case all $a_i b_i > 1$; assign plus if $b_i > 1$ and minus otherwise. The opposite witness cannot include that coordinate in its support. Every nonbase child decreases the nonnegative rank $d + B$: pivot children multiply products by $4/3 \geq \rho_0$, continuation by ρ_0, and the one provisional child keeps masses but reduces d by one. There are at most $2048n + 3$ children decreasing B and at most one preserving it. These observations prove termination and, inductively using the previous paragraphs, simultaneous correctness of all witness claims. Integer validity holds because a nonbase call has $d \geq 12$; the child widths $q = h2^{d-7}$ and $D/2 = h2^{d-1}$ have the permitted form. Each centered child lies between $A + D\mathbf{1}/8$ and $A + 7D\mathbf{1}/8$.

8 From Labeling to the Bit-Model Decision

The source puts

$$N = 32H^2, \quad K = 256(N+1), \quad S_0 = 16HK, \quad J = S_0 W + N + 1, \tag{8.1}$$

and defines $F(t) = S_0 f_{1/N}(t/S_0)$, whose fixed point is $x^* = S_0 v(1/N)$. This S_0 is a scalar, not the synthesis matrix S. The integer comparison vectors are $Y = \lfloor x^* \rfloor - N\mathbf{1}$ and $Z = \lceil x^* \rceil + N\mathbf{1}$. Since $\gamma(N+1) = N - 1/N \leq N$, monotonicity and (7.1) give

$$F(Y) \geq Y, \quad F(Z) \leq Z, \quad Y \leq x^* \leq Z, \quad Z_i - Y_i \leq 2(N+1) = K/128,$$

and $|Y_i|, |Z_i| \leq J$. These vectors are used in the proof only; knowing them computationally would already require locating the fixed point.

8.1 Box invariant and exact output

Start at $A = -J\mathbf{1}$ and $D_0 = 128 \cdot 2^{d_0} \geq 2J$ with minimal nonnegative d_0. For each outer call use $a_i = b_i = 1/n$, so both comparison vectors satisfy every mass restriction. A plus label at i implies $Z_i > A_i + D/32$; a minus label implies $Y_i < A_i + D - D/32$. For $D > K$, choose the next smaller permitted width D': $(h-1)2^d$ if $h > 128$, or $255 \cdot 2^{d-1}$ otherwise. Then $\delta = D - D'$ lies in $(0, D/128]$. Raise the lower endpoint by δ at plus coordinates and lower the upper endpoint by δ at minus coordinates. For plus, using $Z_i - Y_i \leq K/128 < D/128$, the operative estimate is

$$Y_i \geq Z_i - K/128 > Z_i - D/128 > A_i + 3D/128 \geq A_i + \delta.$$

For minus, $Z_i \le Y_i + K/128 < Y_i + D/128 < U_i - 3D/128 \le U_i - \delta$. Both vectors remain inside. There are at most $128(d_0+1)$ outer calls, because each exponent admits 128 mantissas. At termination $D \le K$,

$$|A_i/S_0 - v_i(1/N)| \le K/S_0 = 1/(16H), \qquad |A_i/S_0 - G_i| \le 1/(8H).$$

Proposition 5.3 now gives the source's integer output test $A_i \ge -8K$. Zero is included deliberately; a strict positive-value test would solve a different problem.

8.2 Where the quasipolynomial count comes from

All quantities in (8.1) and the initial box have $O((L+2)^2)$ bits, and $d_0 = O((L+2)^2)$. At integer t, an edge expression for F is $(S_0 w(e) + (N-1)t_j)/N$. For a chance probability c_e/d_e, the common denominator QN yields numerator

$$\sum_{e:i\to j} c_e \frac{Q}{d_e}\big(S_0 w(e) + (N-1)t_j\big).$$

This uses the actual input denominators, also for unreduced fractions and zero probabilities. Exact signed floor, extrema and rational arithmetic have polynomial bit cost. Numerical magnitudes are never unit-cost inputs.

Initially $\mu = 1/n^2$, so $B_0 = O(\log(n+1))$, with $B_0 = 1$ at $n = 1$. Write $\Lambda = d_0 + B_0$ and $s_* = 2048n + 3$. Every path has length at most Λ and at most B_0 branching edges decreasing B; all other edges, if present, have a uniquely specified provisional child. Therefore the recursion-node bound is

$$\sum_{\ell=0}^{\Lambda} \sum_{j=0}^{\min(\ell,B_0)} \binom{\ell}{j} s_*^j \le (\Lambda+1)(B_0+1)(1+\Lambda s_*)^{B_0} = 2^{O((\log(L+2))^2)}. \tag{8.2}$$

Counting a polynomial-depth tree as if it branched at every level would not prove this bound. Each mass-changing edge uses fixed rational factors, and there are only $O(\log(n+1))$ of them on a path. Unreduced mass fractions consequently have polynomial, in fact logarithmic-in-n, bit length. Base iterations have constant maximum numerical width; all other nonrecursive work and the number of outer calls are polynomial. A depth-first stack has polynomial depth and polynomial-size records, so Turing-machine copying/scanning introduces only polynomial factors. This is the source's complete algorithmic mechanism. The independent labeling/application audit is separately attributable; the author does not claim a second implementation replay of it.

Corollary 8.1 (Rational rewards and rational thresholds)**.** *Assume the source algorithm and its proved integer-reward guarantee. For explicit binary rational edge rewards and a binary rational threshold θ, the set $\{i : \mathrm{val}_w(i) \ge \theta\}$ reduces to that guarantee with polynomial encoding expansion. Its resulting bit bound remains $2^{O((\log(\mathcal{L}+2))^2)}$, where $\mathcal{L}$ is the full original input length including the threshold.*

Proof. Write $w(e) = a_e/b_e$, $b_e > 0$, and $\theta = p/q$, $q > 0$. Let $D_w = \prod_e b_e$, and replace the reward by the integer

$$\widetilde{w}(e) = q(D_w/b_e)a_e - pD_w = qD_w(w(e) - \theta).$$

The scalar qD_w is positive. For every play, finite averages transform by this increasing affine map; hence $\mathrm{MP}_{\widetilde{w}} = qD_w(\mathrm{MP}_w - \theta)$. Bounded expectation and the strategy extrema commute with the same map. Thus the transformed nonnegative-value set is exactly the requested threshold set, including equality and nonterminating paths. Since $\log D_w \le \mathcal{L}$ and all input numerator/denominator lengths are included in $\mathcal{L}$, each new reward has $O(\mathcal{L})$ bits and there are $O(\mathcal{L})$ edges. The total new

length is $O((\mathcal{L}+2)^2)$. Exact multiplication and division for these integers have polynomial bit cost, and $\log O((\mathcal{L}+2)^2) = O(\log(\mathcal{L}+2))$, proving the bound. □

This elementary extension was suggested by the independent source reviewer and is credited accordingly. It does not silently replace integer rewards by rationals in the earlier Cramer bounds; those bounds apply to the transformed integer instance. Nor does it assert a new global complexity result independently of the source algorithm.

9 Current Public Context and Version Check

At the recorded public GET check on 8 October 2026 UTC, the repository's `main` resolved to the specified pin, with commit date `2026-10-06T21:58:50Z`. All seven public source/bibliography files returned HTTP 200 from the pinned raw URLs and matched the local bytes exactly. The detailed paths, line counts and SHA-256 values are in `evidence/readscope-public-check.json`. A public repository claim is not by itself a peer-review or formal-verification certificate.

Shapley's contraction framework, Blackwell's eventual optimality principle, and Liggett–Lippman's perfect-information stationary theory are classical antecedents identified by the source. The argument in Sections 3 and 4 was rederived so that the exact payoff convention is not delegated to an ambiguous citation. Andersson and Miltersen's 2009 paper establishes polynomial-time Turing equivalences among these finite stochastic game formulations with binary numerical data [3]; its bibliographic record and primary-paper abstract were checked, not its entire proof.

Gaubert, Grand-Clément and Katz's NeurIPS 2025 paper supplies refined Blackwell and sensitive-optimality thresholds [4]. Its primary paper's assumptions, main results and discussion explicitly distinguish a short description of a discount from algorithms whose running time depends on its inverse gap. Its discussion states that those threshold bounds alone do not yield a quasipolynomial solver. We do not replace the present source's constants by their sharper thresholds without a separate normalization and payoff-model check.

Zwick's public July 2026 preprint, version 1, gives an improved randomized subexponential analysis for turn-based games [5]. This is a different randomness and complexity guarantee, not verification of the October source's deterministic assertion. Azeem, Kretinsky and Weininger's *Sound Value Iteration for Simple Stochastic Games*, public version 2 of 29 March 2026, treats end components in certified value iteration [6]. Its primary version record and abstract were checked; its experimental claims are not transferred to our certificate implementation. Auger, Badin de Montjoye and Strozecki's MFCS 2021 primary publication record provides the established strategy-improvement context [7]. These targeted checks are not an exhaustive novelty search, nor a claim that no later paper exists.

10 Worked Certificates and Assessment

10.1 Strictly sharper localized discounted error

For $u = 1/2$ and

$$P = \begin{pmatrix} 1/2 & 1/2 \\ 0 & 1 \end{pmatrix}, \qquad r = 0, \qquad d = \begin{pmatrix} 0 \\ 1/100 \end{pmatrix},$$

the value is zero and the exact envelope is

$$e = (I - P/2)^{-1} d = \begin{pmatrix} 1/150 \\ 1/50 \end{pmatrix}.$$

Take $x = -e$. Then $f_u(x) - x = d$ and $v(u) - x = e$ exactly. The scalar bound $\|d\|_\infty / u = 1/50$ is three times larger at state one. This is a genuine improvement over the coarse residual norm, not over the optimal classical resolvent. In the one-sided policy repair, overestimates only need positive lower defects and underestimates only need positive upper defects, so absolute values can discard useful sign information. For example a single reward-zero loop at $x = -1$ is already a valid lower bound with $d^- = e^- = 0$, whereas its symmetric envelope is one. Both statements follow by exact substitution.

10.2 A nonconstant-gain certificate

Let Max at s choose a reward-zero edge to a or b; let a, b be absorbing with rewards $+1, -1$. The value and small-discount bias are $g = (1, 1, -1)$ and $b = (-1, 0, 0)$ when Max chooses a. All tests in Theorem 4.3 hold with $\varepsilon = 0$: the selected edge has bias expression one, while the alternative Max edge decreases gain and need not pass a gain-preserving bias test for the upper certificate. At a, b the loop identities are exact. This illustrates why demanding one common scalar gain would exclude legitimate multichain games. In the chance version with equal chance of a and b, use $g = (0, 1, -1)$, $b = (0, 0, 0)$. Gain is harmonic at s, then changes once at most; the expected-liminf value at s is exactly zero.

10.3 Contribution score, advantages, and costs

The supported score is **C2/5**. The local advance is an auditable source-specific synthesis: asymmetric discounted repairs, gap-aware zero classification, bias-defect behavioral witnesses, a gain-change count, and a sharp stopping-sensitive discount constant. Each has a complete proof and testable rational conditions. Classical contraction, resolvent, martingale, LP and determinant methods are credited; no historical novelty or independent discovery of the source's recursion is asserted.

Advantages are tighter localized errors, sound inclusion of zero, explicit opponent quantifiers, and inexpensive checking once a rational certificate is supplied. Costs include obtaining that certificate, exact-arithmetic growth, potentially loose universal H, and LP work for envelopes. The gain-change bound gives no useful universal clock bound; a stopping drift can have exponential numeric size; a reconstruction can have a large full residual despite a small projected one. The report supplies neither a performance benchmark nor a general asymptotic improvement over the source.

10.4 Concrete future research

1. Compare the componentwise and one-sided certificates on exact solver outputs, measuring full bit cost and rejected actions rather than only floating-point residual size.

2. Derive model-compatible sharper gain separation and discount constants from the public Blackwell literature, preserving edge-reward timing and the expectation-of-liminf convention before substituting constants.

3. Develop exact sparse verification of g, b and chance harmonicity, with a proved witness-discovery bound if a complexity improvement is claimed.

4. Formalize the labeling translation, stopping-pivot mass inequalities, and ranked tree count as separate lemmas. Finite testing alone cannot prove their universally quantified witness guarantee.

5. Explore stopping-region decompositions with a declared zero boundary; refuse a global residual conclusion whenever its required drift or exact gain tests fail.

A Read Scope, Exact Tests, and Bounded Validation

The public source read scope comprises 69 lines of main source, 197 of introduction, 378 of discounting, 608 of labeling, 278 of application, 42 of the figure, and 158 of bibliography. These counts refer to the byte-verified pinned files. The actual private passages read are recorded with their local path and SHA-256 in the read-scope evidence; no whole corpus audit is claimed. The certificate results stand on the proofs here and do not require accepting an unexamined foundational transform theorem.

The companion `evidence/author_checks.py`, executed using local Python 3.10 and SymPy, passed 5,670 exact assertions. It covers 48 seeded mixed games (32 with three states and 16 with four), evaluating all 224 positional pairs by an independent recurrent-class/stationary-average calculation and transient rational solve. The selected small-discount pair is checked against that baseline, with exact symbolic gain and bias, all opponent-edge conditions, and a controlled bias perturbation. It also checks all 75 one-state two-reward arenas over rewards $-2, \dots, 2$ and three ownership types at two discounts, the sharp localized envelope, nonstopping counterexamples, zero margins and rare-escape probabilities 2^{-b} for $1 \le b \le 128$. An additional 100 rational-reward/threshold preprocessing cases and 128 small-gain-drift counterexamples check the added corollary and the exact-harmonicity boundary. Counts and exact game data are retained in `evidence/mathtests.json`. The source's recursive algorithm is not reimplemented by the author. Finite results do not certify literature priority, all game sizes, all behavioral strategy spaces by enumeration, or a speedup. The complete conditional-expectation proofs, not a sampling experiment, cover behavioral opponents.

A.1 Independent and parent mathematical reviews

The independent reviewer read all seven actual public files, focusing on the complete labeling/application chain. The bounded audit found no concrete counterexample, incorrect inequality, nonterminating branch, or missing bit-complexity factor. It emphasized that the finite-box fixed point need only be a subsolution; that the last unchanged pivot call is indispensable; that the mass budget is counted along paths; and that the N-padding and S_0 scaling explicitly pay for inverse-discount amplification. It also checked expected-pathwise-liminf semantics and exact zero. This is the outcome of an error search, not a global validity certificate or formal proof.

The review's independent standard-library structural script reports 7,807 width predecessor cases, 131,200 pivot-grid positions, 48 clipping boundary cases, 2,451 simultaneous peak-protection updates, 4,480 mass assignments, 577 rank inputs, 65 exact rank-boundary cases, and 1,984 recursion-tree triples, all passing. These are the reviewer's tests, not an additional author replay. Copies of that review and script are retained as immutable evidence snapshots with separate hashes. They do not test the new author certificates; those are covered by the proofs and author suite above.

The parent also read the complete assembled mathematical report, the actual private stopping/residual passages, and the full source discount section, and reported no decisive mathematical defect. One exposition issue was identified and corrected: the box-gap premise is the non-strict $Z_i - Y_i \le K/128$, followed by the strict $K/128 < D/128$, exactly as used in the displayed containment proof. The rational-reward extension retains its separate source-dependent status. These bounded reviews do not imply an exhaustive all-input or formal verification.

A.2 Native and physical build evidence

The native editor open was attempted and returned a queued result. Native compilation was attempted and failed with the diagnostic `Unabletofindstandarddirectoriesforplatform`. This is an environment limitation, not a successful compilation claim. The pre-existing physical exporter

uses the installed TeX engine and bundled PDF tools; no software was installed. Build logs and rendered-page evidence accompany the delivered files, and the frozen manifest identifies the exact TeX/PDF bytes, mathematical tests, read scope, and review snapshots. Shared mutable catalogues and progress files are not part of that immutable evidence set.

References

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